

Can an AI agent solve a PDE and prove it?

Qwen3-8B on AWS Trainium2

OUR IDEA

*Generate a mathematical answer. Check it. Explain the error.
Try again.*

THE TEAM

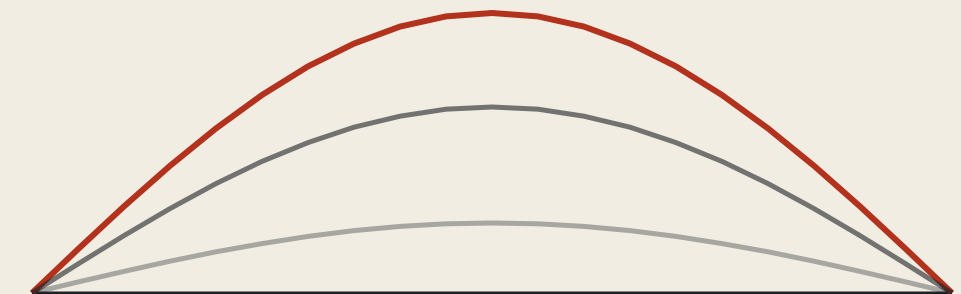
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THE PROBLEM

A formula can look correct and still be wrong

Our task: predict temperature across a rod as heat diffuses over time.

PDE residual

Does it obey physics?

Boundary values

Are the ends correct?

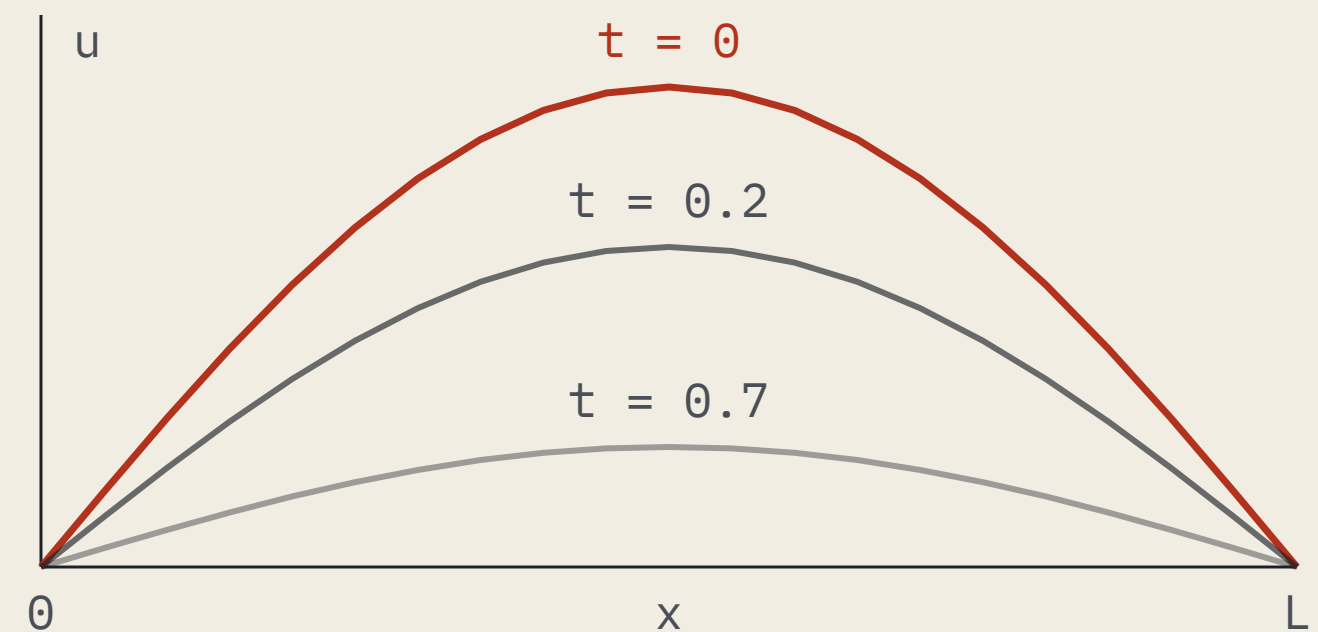
Initial condition

Does it start correctly?

THE PHYSICS

Heat equation	$u_t = a u_{xx}$
Endpoints	$u(0, t) = u(L, t) = 0$
Starting profile	$u(x, 0) = f(x)$

ILLUSTRATIVE HEAT DIFFUSION



A correct solution cools smoothly while satisfying all constraints.

OUR APPROACH

The agent is a loop, not a one-shot chatbot

Qwen3-8B proposes; tools calculate; a checker scores and explains failures.

01

TASK

Heat equation
+ conditions

→ 02

MODEL

Qwen3-8B
Trainium2

→ 03

TOOLS

Calculator
+ SymPy

→ 04

CHECK

PDE + boundary
+ initial

→ 05

REVISE

Select candidate
or try again



FEEDBACK LOOP

Failed PDE residual? Incorrect boundary? Wrong Fourier coefficient?

Give the model a specific mathematical diagnosis for its next attempt.

A verifier that checks the math, not the wording

Our symbolic checks provide structural evidence beyond fluent-looking text.

01

PDE residual

Compute $u_t - a u_{xx}$; should be zero.

Does the formula satisfy diffusion?

03

Initial condition

Compare $u(x, 0)$ against $f(x)$.

Does it match the given start?

02

Boundary conditions

Evaluate $u(0, t)$ and $u(L, t)$.

Are endpoint temperatures valid?

04

Fourier coefficients

Check mode weights when applicable.

Are the series terms consistent?

See the error before accepting the answer

Illustrative example: the wrong decay rate breaks the heat equation.

✓ CORRECT MODE

$$u(x,t) = \sin(\pi x) \exp(-a \pi^2 t)$$

Residual $u_t - a u_{xx} = 0$

Same spatial wave; matching decay rate.

The symbolic checker can validate this.

Demonstration of the verification principle; not an observed benchmark trace.

⚠ WRONG DECAY RATE

$$u(x,t) = \sin(\pi x) \exp(-\mathbf{2}a \pi^2 t)$$

Residual is nonzero

The equation fails even though the boundary and initial values look fine.

ACTION: revise exponent

MEASURED RESULTS

Verifier-guided runs: 5 out of 5 solved

12 recorded runs across three variants; these are small, non-matched samples.

VARIANT	RUNS (FILLED = SOLVED)	SOLVED	MEAN BEST REWARD
Baseline		4/5 (80%)	0.96
Verifier-guided		5/5 (100%)	1.00
Independent selection		2/2 (100%)	1.00

Caution: stochastic runs; different variants use different revision/selection calls.
Observed results do not establish statistical significance or a controlled speedup.

THE TAKEAWAY

A small model becomes more reliable with feedback

What we delivered, what we learned, and what should be tested next.

BUILT

Qwen3-8B on Trainium2

Math tool + SymPy verifier

Candidate revision/selection

OBSERVED

Baseline: 4 / 5 solved

Verifier-guided: 5 / 5

Independent selection: 2 / 2

NEXT

Matched-seed evaluation

Cost / latency per solve

More PDE families

Core idea: do not trust the first formula - verify constraints and use the errors to improve it.

REPRODUCIBLE SUBMISSION

github.com/yahavb/trainium-agent-labs/pull/11